

Important Directions

- **Read through all slides**
 - **Watch all videos**

Infancy

Physical Development

Principles of growth

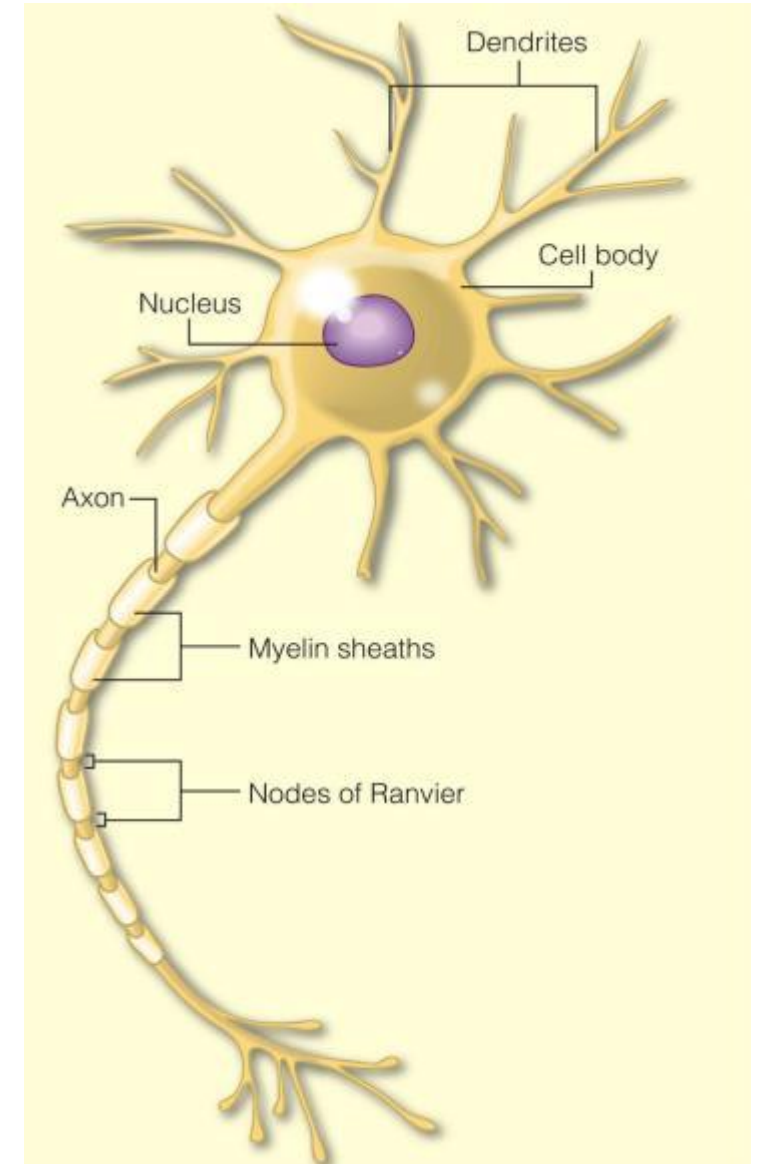
- Sequence of maturity
- Cephalocaudal Development
 - “head to foot”
- Proximodistal Development
 - “center to periphery”

Why does the head begin to grow first?

- Humans have large brains, and hence large heads that need to develop early and continue to develop for an extended period of time.

Neuron

- Cell body (or Soma)- contains nucleus
- Axon- carries messages away from cell body to other cells
- Dendrites- receive messages and directs them to cell body
- Myelin sheaths- fatty substance that supports brain cells
- Synapses- the spaces between the **dendrites of one neuron and the axon** of another where chemical messages are exchanged. This is how neurons communicate.



Transient exuberance

- At birth, the brain has far more neurons than one will ever use BUT far fewer synapses than one will ever use.
- Great increase in the number of dendrites that occur in an infant's brain in the first two years of life.
- Temporary- Pruned by experience

Selective Cell Death

- Even as the brain gets *larger* with development, the number of neurons actually decreases.
- Brain overproduces at the beginning of life, then **prunes away neurons that are not activated by sensory and motor experience**
- This allows the environment, hormones, and genetic signals to shape the brain.
- “Use it or lose it”

Plasticity

- Sensitivity/Responsiveness to the environmental experiences
- Think of resilience/flexibility/ability to bounce back
- Plasticity important for the infant's brain development-
- Positive and Negative implications
- Brain capable of benefiting from good experiences BUT also capable of being harmed by bad experiences
- Different areas of the cortex are equipotential.
 - Neurons can assume functions of lost neurons
 - Neurons are plastic- but more limited as you age.
- Typically, the older the brain the less plastic.

The Romanian Adoptions

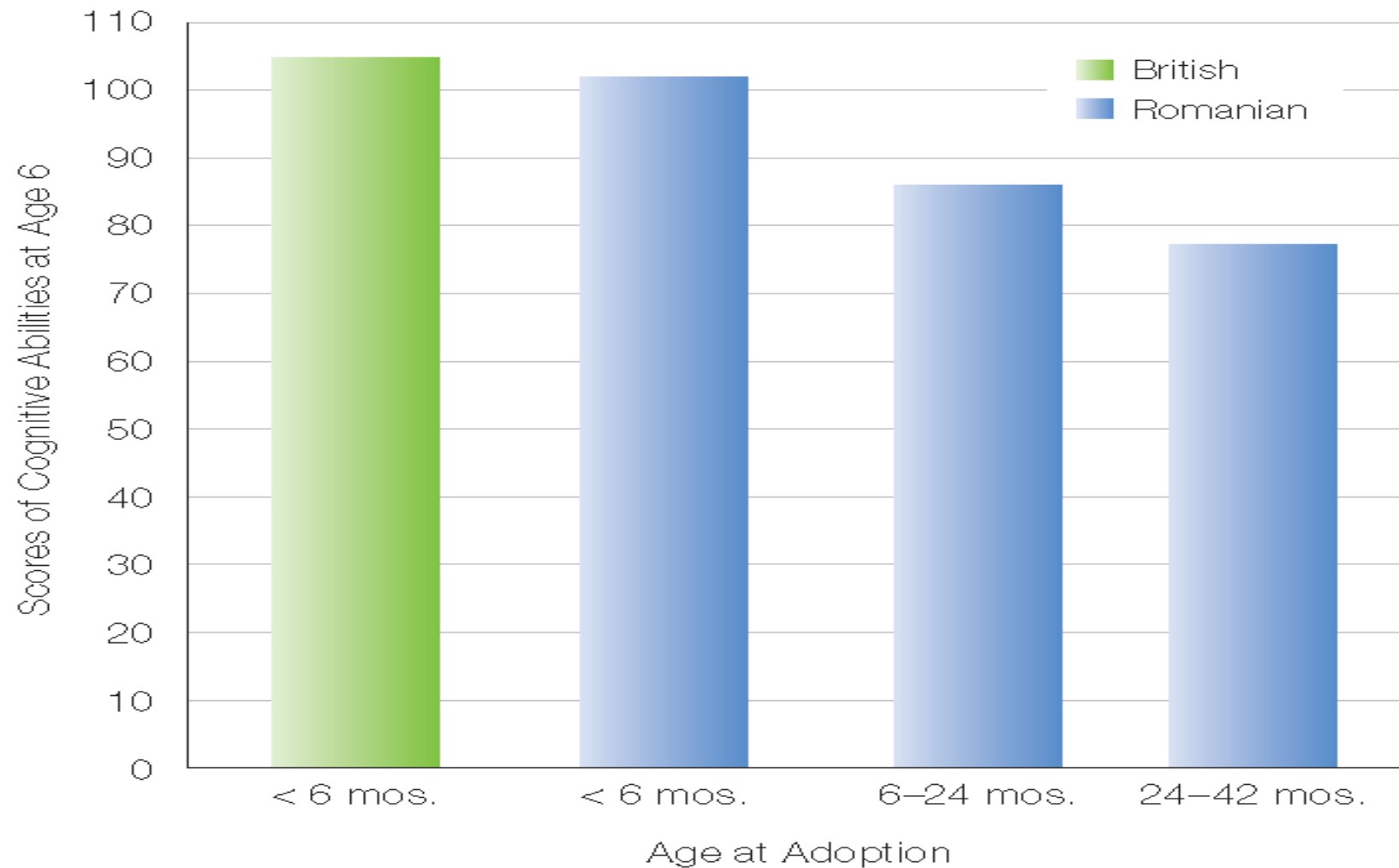
Early Environmental deprivation can have permanent ill- effects

Example seen in Romanian adoptions:

<https://www.youtube.com/watch?v=ptZ-L3uAXXA>



Romanian Adoption (Beckett et al., 2006)



Why does institutionalization have this effect?

Experience Expectant Synaptogenesis

- Young brains have evolved to expect species typical experiences- sensory stimulation, nutrition, and social stimulation.
- In the absence of such stimulation, selective cell death may go awry, such that too many neurons and synapses are lost without replacement.
- Another good example: Language

<https://www.youtube.com/watch?v=6H2POnmzbPo>

Experience Dependent Synaptogenesis

- Synapses that require specific environmental experiences
- Lead to idiosyncratic abilities in individuals
- Example: Musical abilities

Sleep changes: Sudden Infant Death Syndrome

- Infants between 2 to 4 months of age have highest risk for Sudden Infant Death Syndrome (SIDS)
 - Leading cause of death for infants 1 to 12 months in developed countries
 - Sleeping on stomach instead of back
 - Low birth weight and APGAR score
 - Maternal smoking
 - Soft bedding
 - Solution: Sleep ON Back

Co-sleeping?

- Think about co-sleeping with your infant.
- What do you think about it? Do you plan to co-sleep with your virtual child?
- What are your reasons? What are your concerns-both for and against it?
- How have these reasons been shaped by your cultural background?
- Culture plays a big role
 - Is independence emphasized?
 - Is collectivism emphasized?
 - Findings
 - No emotional harm caused by co-sleeping
 - Children who co-sleep as infants grow up to be more self-reliant (can dress themselves) and socially independent (make new friends easily) Keller & Goldberg, 2004

What about SIDS?

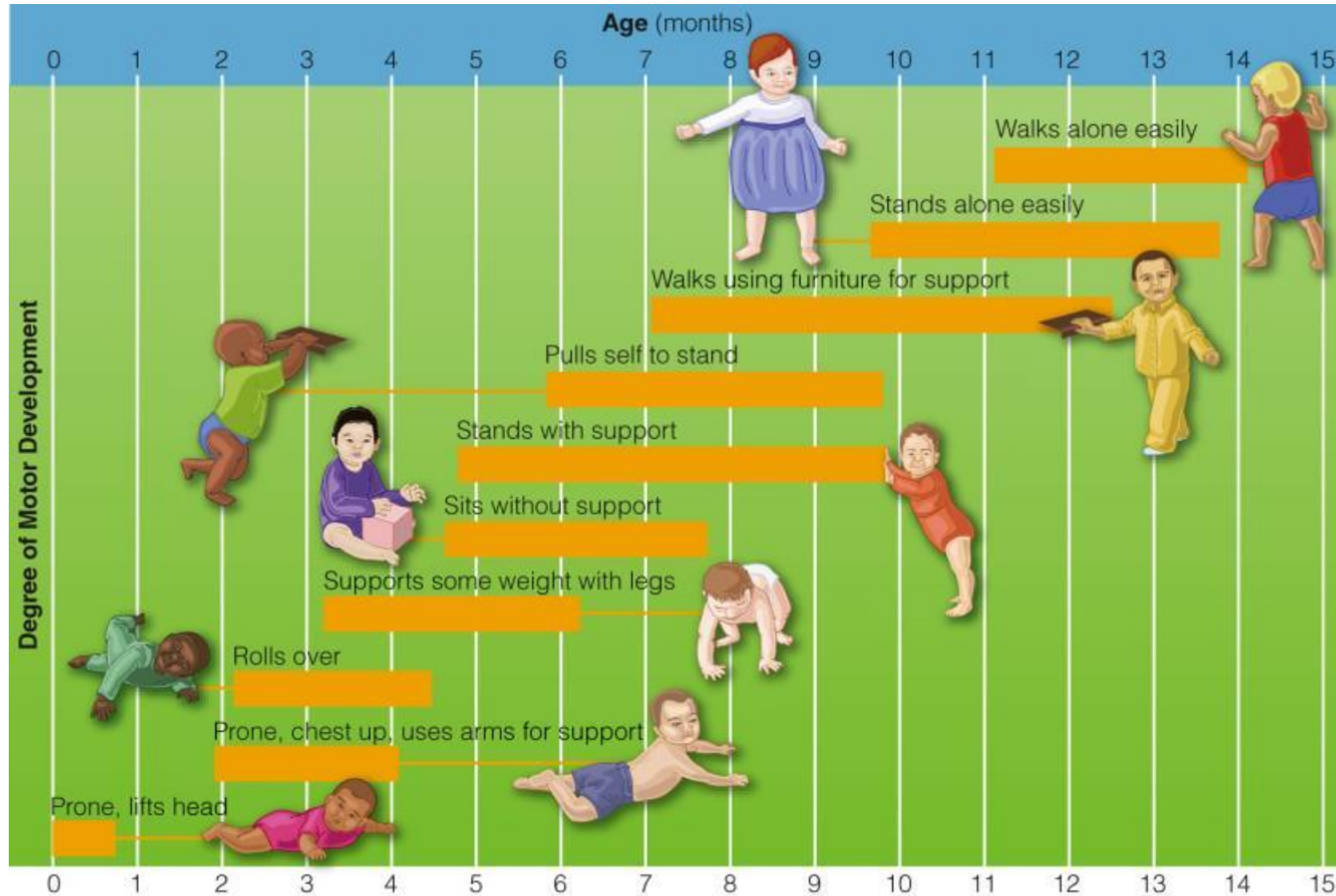
- SIDS is almost unknown in cultures where co-sleeping is the norm
- In fact co-sleeping is a **protective factor** against SIDS
- WHY?
- Parents and infants likely to sleep on hard surfaces
- Co-sleeping encourages breast-feeding- mother more vigilant at night, also infant likely to sleep on back for easier nursing.

Motor Development

- Gross Motor Development
 - Motor behaviors associated with large muscles
 - Crawling, walking,
- Fine Motor Development:
 - Mostly use of the hands
 - Picking items up, writing, etc.



Some developmental milestones



- How can we assess infant sensory perception?

Visual Preference Paradigm

- We measure the amount of time infants spend looking at different visual stimuli
- Differences in looking time evidence the ability to discriminate stimuli.

Infant Visual Preferences

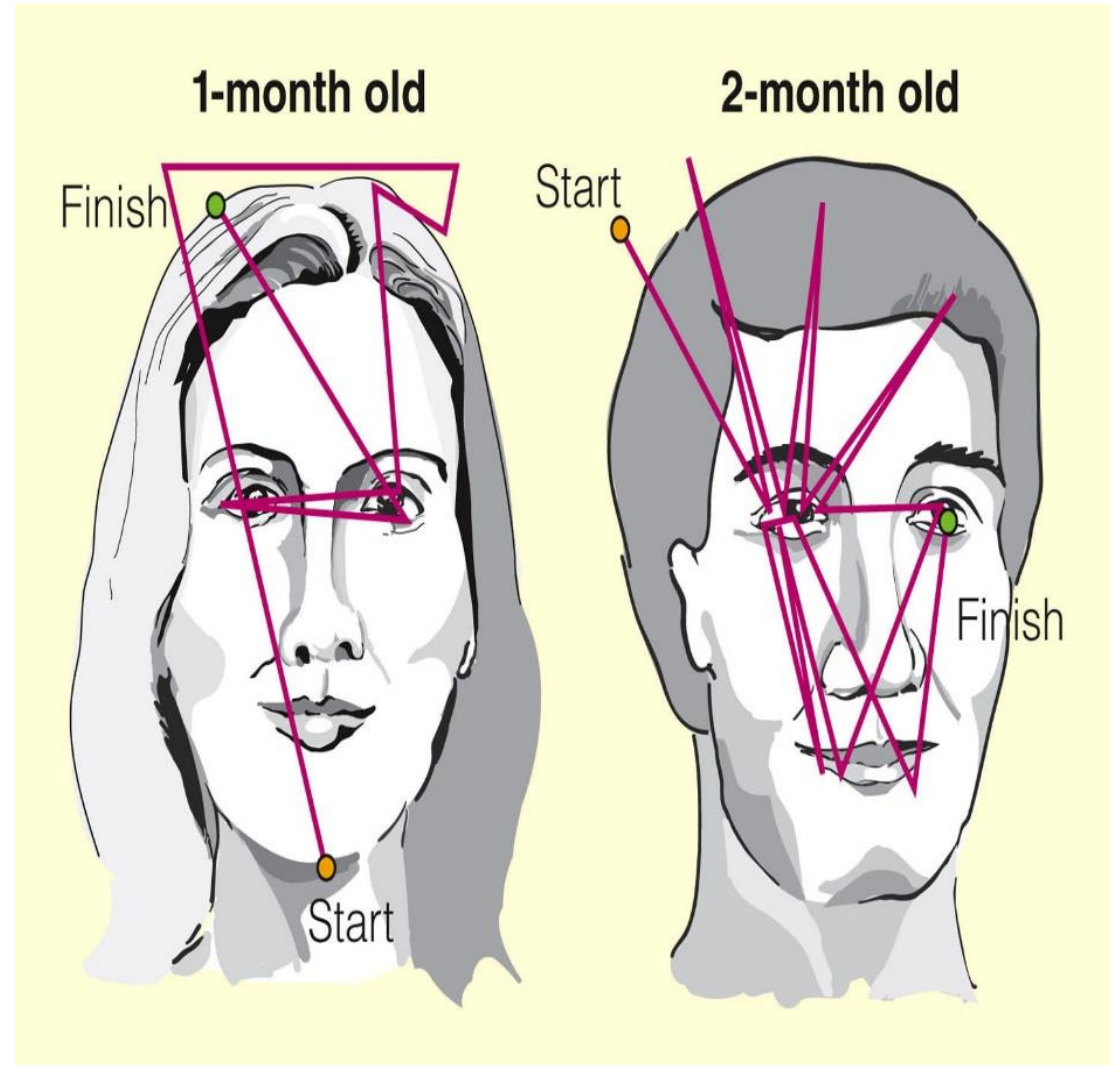


Infant Visual Preferences

- Perceptual biases: prefer to look at some things more than others
- Physical: movement, complexity, high contrast, vertical symmetry, curved lines
- Faces are special

Face Processing Development

- At 1 month, infants look at the outside of the face, *externality effect*
- But by 2 months, infants focus on internal features, specifically the eyes!
- By 8 months, they shift attention to mouth- around the time they learn language



Speech Perception

- Babies are able to tell the difference between *phonemes*, the smallest unit of sound in language
 - Babies listened to “*pa*” while sucking a pacifier, until they decreased their rate of sucking
 - When presented with “*ba*” they begin sucking on the pacifier at a high rate again.
- Apparently, young infants can tell the difference between *all phonemes in all languages*.
- But, *perceptual narrowing* begins and babies lose this ability and focus on their *native language around 6-8 months*

The linguistic genius of Babies: Watch TED talk

https://www.ted.com/talks/patricia_kuhl_the_linguistic_genius_of_babies

Test your understanding: Answer these questions from the TED talk. You do not have to turn these in, however these are important from the exam perspective.

- 1) Why are babies called citizens of the world?
- 2) Unlike babies, adults are _____ listeners.
- 3) When do citizens of the world turn into language bound listeners?
- 4) Around 6-8 months of age, babies from Tokyo and U.S distinguish between la/ra sounds equally well.
True/False
- 5) At 10 months of age, babies from U.S are getting a lot better at distinguishing la/ra sounds but not Japanese babies. Why is this?
- 6) What is the function of babies' being able to distinguish between sounds of all languages?
- 7) Are babies able to take statistics of the language they hear via a TV? What implication does this finding have for educational media meant to teach infants language skills?

Cognitive Development: Infancy

What do infants know? When and How do they know it?



Piaget's Cognitive Developmental Theory

Sensorimotor Period (birth-2)

- Thinking is tied to physical action. Infants come to know the world by their actions.

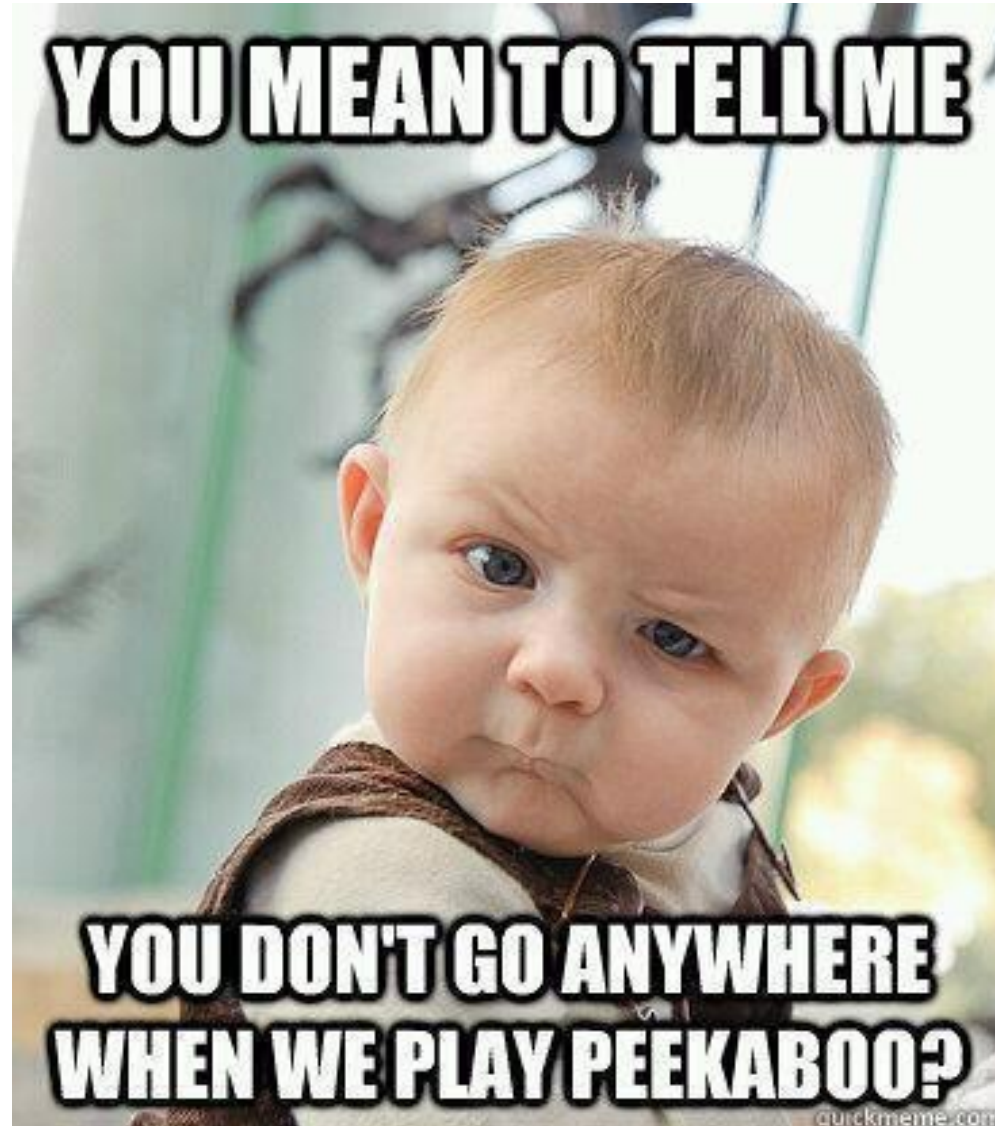


Sensorimotor Period

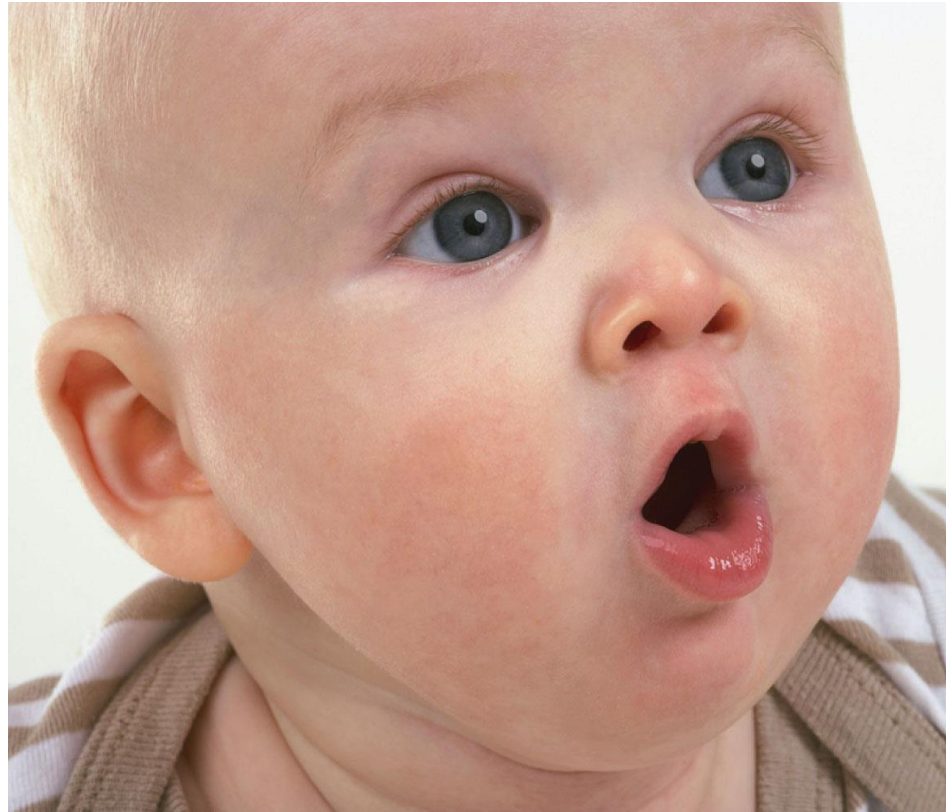
- Has 4 stages:
 1. The Use of Reflexes
 2. Primary circular reactions
 3. Secondary circular reaction
 4. Co-ordination of secondary schemes



During sensorimotor period, they achieve Object Permanence



- How can we know what babies think and know?



Procedure: Violation of Expectation Paradigm

Based on the principle that **looking time** increases when infants view something that violates what they expect to see

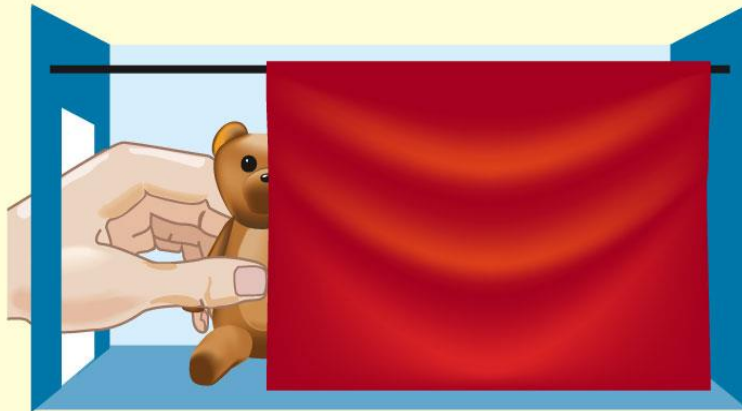
(Wynn, 1992) Violation of Expectation and Infant's Understanding of Number

1. Object placed on stage 2. Curtain comes down

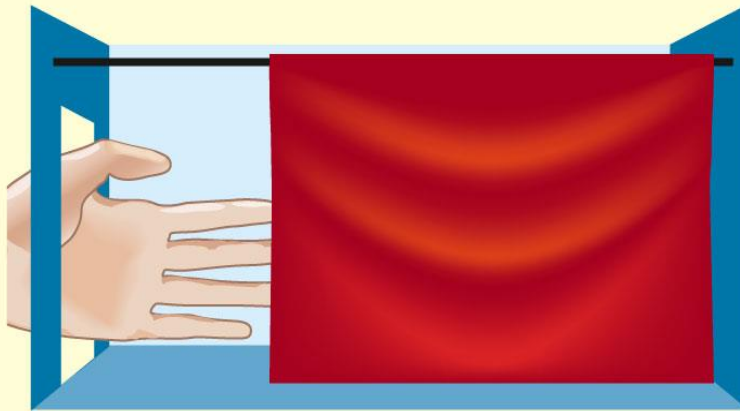


(Wynn, 1992) Violation of Expectation and Infant's Understanding of Number

3. Second object added



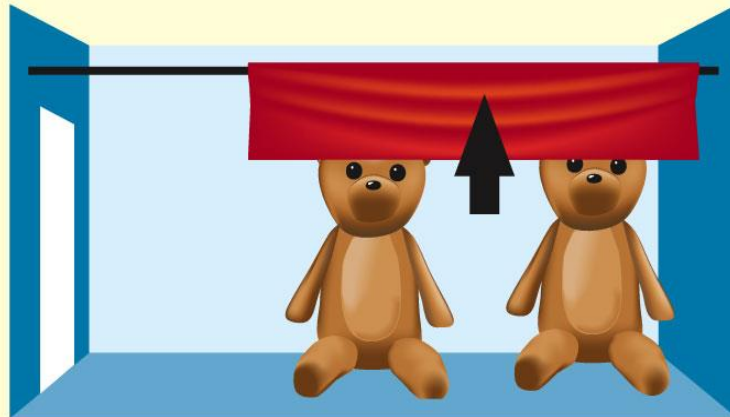
4. Hand leaves empty



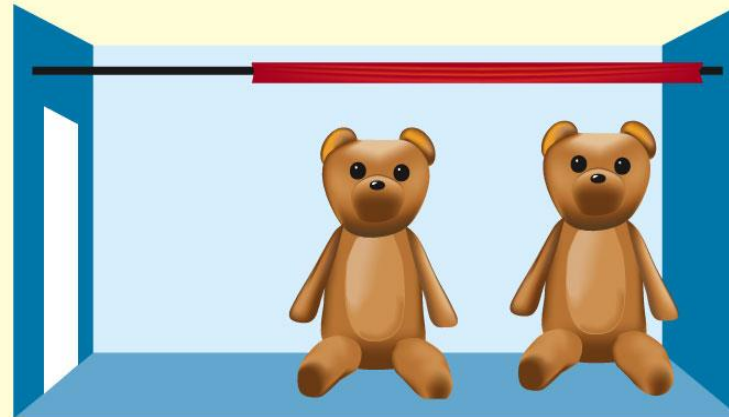
(Wynn, 1992) Violation of Expectation and Infant's Understanding of Number

Either possible outcome

5. Curtain goes up



...revealing two objects



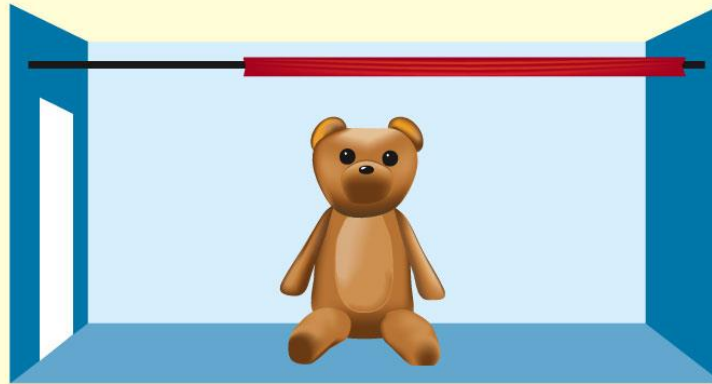
(Wynn, 1992) Violation of Expectation and Infant's Understanding of Number

Or impossible outcome

5. Curtain goes up



...revealing one object



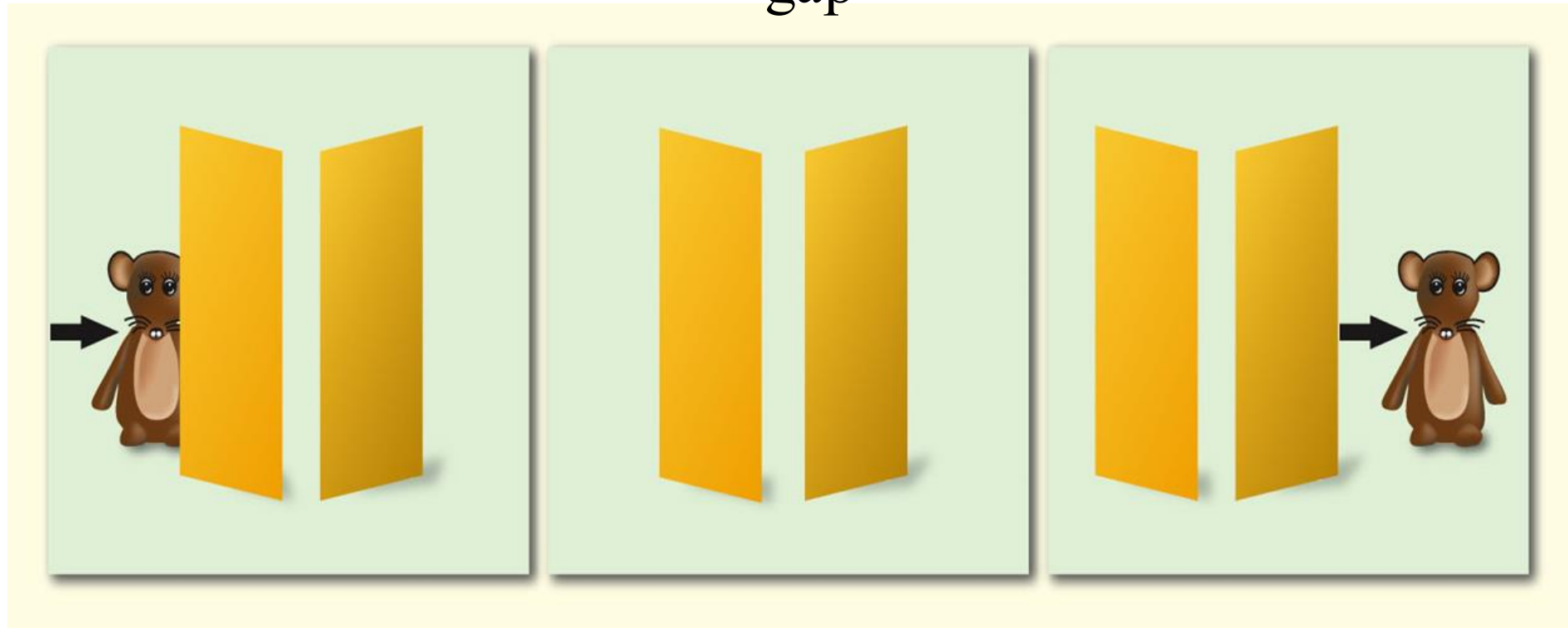
For the impossible outcome, infants show surprise, and an increase in looking time



“Now, wait a minute, that DOES NOT look right...???”

Objects: Permanence (Baillargeon, 1987)

Infants at 2.5 months know that the mouse should appear in the
gap



<https://www.youtube.com/watch?v=l1VK2iawS34>

Habituation/Dihabituation Paradigm

- Habituation:
 - Repeated exposure=Decreased looking time.
 - Infants get habituated to this stimulus.
- Dishabituation:
 - Presenting new stimulus = increased looking time

“Oh Wow...This is some kind of cool new thing!! Let me look at it intently!”



“Hmm. Now I am bored. I have looked at this for long enough!”



Habituation: Decrease in looking time

Dishabituation

- If you show the infant a novel thing after they have habituated to an old stimulus, there is an increase in looking time, again.

What does this tell us?

- Infants are discriminating between the two stimuli.
- Habituation/dishabituation shows basic of recognition memory.

Media and Learning in Infancy???

You notice that Indeera (My Virtual Daughter) can learn a thing or two so you start working on her baby IQ!

What do you think I should do?



A) You show her lots of educational DVDs especially designed for infants

B) You show Indeera interesting objects such as rattles, mobiles, and stuffed toys and watch how interested she is in looking at them or reaching for them.

C) You talk a lot to Indeera even though you know she probably doesn't understand you. You hope this will promote language skills later on.

Media and Learning in Infancy-Findings

- Educational Media Products do NOT improve infant cognitive development.
- In one study, with every hour of educational DVD watching among 8-16 month old infants, infants understood 8-16 fewer words (Guernsey, 2007)
- WHY??
- Human interaction is important input for language development.
- Effective ways to encourage cognitive dev- talking, reading, responding



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Language Development

- Around 2-4 months infants begin “cooing”
 - “oooooh” noises
 - Before this only reflexive sounds (crying, sneezing, etc)

MVC: You love to hear Indeera make those little "ooo" sounds.

- ★ 1) Give her food to increase cooing
- ★ 2) Spend a lot of time talking to her to encourage more sounds
- ★ 3) You think cooing is to get your attention- so you try to give her attention every chance you get.



Babbling begins around 6-10 months

Repeated sequences of consonants and vowels

“nanananana” or “bababababa”

Even hearing impaired infants “babble” with their hands
(Petitto, 2002)



Emotional and Social Development Infancy

Temperament

- Personality in infancy
- Biological foundation
- Affective, Attentional, and Activational core of personality

Temperament

- 3 Types (Thomas & Chess, 1977)
 - Easy
 - Difficult
 - Slow-to-Warm-Up

Easy Babies (40%)

- ☐ Regular patterns of eating and sleeping
- ☐ Generally positive mood
- ☐ Eager to approach people and objects
- ☐ React to events with low to moderate levels of intensity



Difficult (10%)

- ☐ Unpredictable eating and sleeping
- ☐ Negative moods
- ☐ Difficulty adjusting to new environment
- ☐ React to events with high levels of intensity

Slow-To-Warm-Up (15%)

- Slow pattern or reaction
- Tendency to withdraw in novel situations
- Low in activity

Your Virtual Child: At 12 months of age

- Five dimensions used to describe temperament
- These dimensions are random at birth,
- Are influenced by your questionnaire responses
- Change gradually over time in response to events and your parenting decisions.

MyVirtualChild Temperament Dimensions

- Activity-Physical and Mental activity of child
- Sociability- Friendliness and desire for social interactions
- Emotionality- Intensity and fluctuations of emotions
- Aggressive/cooperativeness in social situations, parental demands
- Self-control

Goodness of fit

- Children develop best if there is a good fit between the temperament of the child and the environmental demands
- Tendency of parents to adapt her/his behavior to the child's temperament.
- For an active child, go for a walk and talk about the bugs you see
 - Instead
- Of quieting him down and making him read a book about bugs
- Difficult and Slow-to-warm up babies need parents to be
- Patient and Nurturing

Indira is a very active crawler and is always on the go. You are afraid she might get into something and get hurt. How do you monitor her activity?



a) Install baby gates everywhere and restrict her movement around the room

★ b) Baby-proof the entire house and let her move freely

★ c) You do not modify the house but always keep her within sight, thinking babies need to explore their natural surroundings.

Goodness of Fit

- The principle of goodness of fit, and the related concept of moderate novelty helps change the child's temperament, or develop a particular skill
- Introduce **moderately novel** activities and experiences, because **children are more likely to pay attention to and profit from such experiences.**

Indira tends to be slow to warm up to new people, but is outgoing and friendly with familiar people.

To make her more comfortable with unfamiliar people



★ a) You introduce her to new situations in small doses, making sure she has time to get used to the new people

b) You introduce her to new people and situations as much as possible, since the stimulation is good.

c) You keep her at home with familiar people, figuring she will grow out of her shyness in time.

★ d) You take her to the same play group often so she can build relationships with other children and adults.

Indira's language skills are developing rapidly, and she wants to talk to you almost all of the time. Indira seems to know an unusual number of names for things. Sometimes you hear her talking to herself as she plays alone in her room.

- A) You are worried that she is talking to herself, so you interrupt her monologue and strike up a conversation
- ★ • B) You have lots of conversations when you hang out together doing routine things like feeding, bathing etc.
- ★ • C) You often look at picture books and talk about them. Indira wants to know the name of everything!



Emotional expression: Primary Emotions

- Emerge during the 1st year of life
- Disgust, surprise, joy, anger, sadness, and fear
- Hypothesized to be “innate” and part of our evolutionary history
 - Therefore should be found across cultures and develop roughly at the same time for all children

Emotional Perceptions

- ☐ Babies are sensitive to others emotions
 - ☐ Contagious crying
 - ☐ Imitation of facial expressions
 - ☐ Sensitive to mother's reactions- specifically smiling/motherese
 - ☐ https://www.youtube.com/watch?v=YTTSXc6sARg&ab_channel=Mindyourclass
 - ☐ Social referencing- how to react
 - ☐ <https://www.youtube.com/watch?v=p6cqNhHrMJA>



Indira learned to walk early (9 months), and even knows how to climb. She is constantly into everything! When she does something dangerous, such as running into the street or climbing onto the back of the couch, you:

- A) Scold her angrily to teach her a lesson about dangerous places
- B) Say “No!”, remove her from the place and put her in a time-out place so she can calm down.
- ★ • C) Put on a scared face, speak in a frightened voice and tell her “No, you can get hurt.”
- D) Spank her so she learns a lesson about dangerous places.



Emotional Regulation

- Newborns have no control over emotions, but simple forms of regulation develop quite quickly
- At 3 months, they can soothe themselves by sucking on a pacifier and withdraw from over-stimulation
- Begin to use adult references for emotional context around the age of 1 (social referencing)

When strangers come to visit, they think Indira is really cute so they want to hold Indira. Indira usually starts crying if the person tries to get too close. However, after meeting a person two or three times, you notice that Indira is staring at the person and manages a shy smile. Typically you:

- A) Let the new person hold her before Indira can change her mind.
- ★ • B) Stay close to her and let her gradually get used to the person and make her choices about interacting with them.
- ★ • C) Smile and converse with the person and encourage Indira to be friendlier, but not put any pressure on Indira.



The foundation of Social Development: Two Theories

- Erikson's Theory
- The psychosocial task of infancy : Developing Trust versus Mistrust
- Is the world a safe place?
- Based on the responsiveness of caregiver: Will others in my life be trustworthy?
- Am I worthy of being loved?

Attachment- Infant-Caregiver Bond

- Especially close **emotional links** established between a child and its primary caregiver (usually the mother)
- Motivational system for
- **Protecting** children from danger
- **Providing care** as they develop

What could attachment be necessary for?

Protection

Exploration

Caregiver acts as a Safe haven

Caregiver acts as a Secure Base

Attachment Theory

- **John Bowlby**

Psychiatrist who worked with infants during and after WWII.

- Mother-raised infants

0% mortality

Orphanage-raised infants

37% mortality

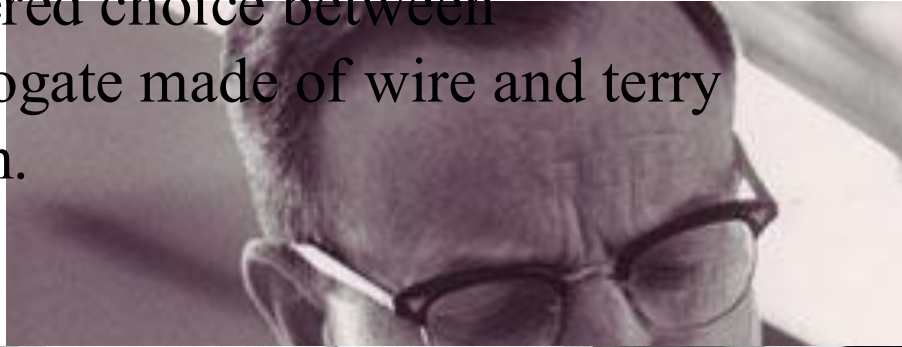
in spite of adequate
medical care

(Spitz, 1945)



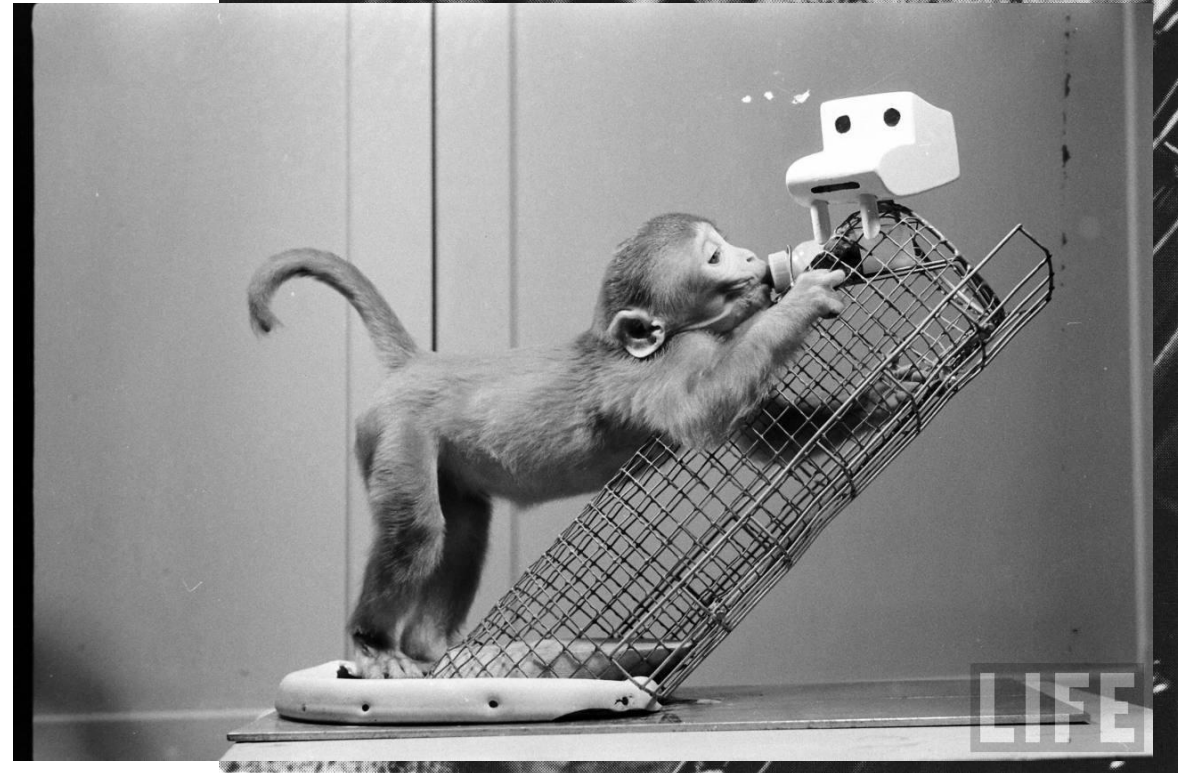
Harry Harlow's series of experiments on maternal separation in rhesus monkeys (1957- 1963)

- Offered choice between surrogate made of wire and terry cloth.



Harlow & Zimmerman, 1959

- The monkeys preferred the terry-cloth mother whether she provided food or not.
- The monkeys chose the wire mother only when she provided food.
- In a day, the monkey spent 17- 18 hours on the cloth mother, less than 1 hour on the wire-mother.
- <http://www.youtube.com/watch?v=OrNBEhzjg8I>



Development of Attachment: Specific attachment

- Separation anxiety
- Wariness of Strangers
- Secure base

Goodness of Fit also applies to Developmental level

- Tendency of parents to adapt her/his behavior according to the child's developmental level

Indira is showing some fearfulness around unfamiliar people and clings to whoever is holding her. What is your typical response?

- ★ a) Ignore it as this is a normal reaction for 8 month olds
- b) Persist in trying to get her to be smiling and friendly if this is an important person
- ★ c) Hold her, and let her warm up to the stranger, if she doesn't get friendly after 10 minutes, try again later or give up.



Goodness of Fit also applies to Developmental level

- Most 6-8 month olds are anxious around strangers
- So you want to introduce them to a new person gradually, and not thrust them to the stranger.